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CANDIDATE NUMBER

SYDNEY GRAMMAR SCHOOL



2014 Trial Examination

FORM VI

MATHEMATICS 2 UNIT

Friday 1st August 2014

General Instructions

- Reading time — 5 minutes
- Writing time — 3 hours
- Write using black or blue pen.
- Board-approved calculators and templates may be used.
- A list of standard integrals is provided at the end of the examination paper.

Total — 100 Marks

- All questions may be attempted.

Section I — 10 Marks

- Questions 1–10 are of equal value.
- Record your solutions to the multiple choice on the sheet provided.

Section II — 90 Marks

- Questions 11–16 are of equal value.
- All necessary working should be shown.
- Start each question in a new booklet.

Collection

- Write your candidate number on each booklet and on your multiple choice answer sheet.
- Hand in the booklets in a single well-ordered pile.
- Hand in a booklet for each question in Section II, even if it has not been attempted.
- If you use a second booklet for a question, place it inside the first.
- Place your multiple choice answer sheet inside the answer booklet for Question Eleven.
- Write your candidate number on this question paper and submit it with your answers.

Checklist

- SGS booklets — 6 per boy
- Multiple choice answer sheet
- Candidature — 91 boys

Examiner

MLS

SECTION I - Multiple Choice

Answers for this section should be recorded on the separate answer sheet handed out with this examination paper.

QUESTION ONE

What is the gradient of the line $6x + 3y - 2 = 0$?

- (A) 2
- (B) -2
- (C) $\frac{1}{2}$
- (D) $-\frac{1}{2}$

QUESTION TWO

What is 5.29784 correct to three significant figures?

- (A) 5.29
- (B) 5.297
- (C) 5.30
- (D) 5.298

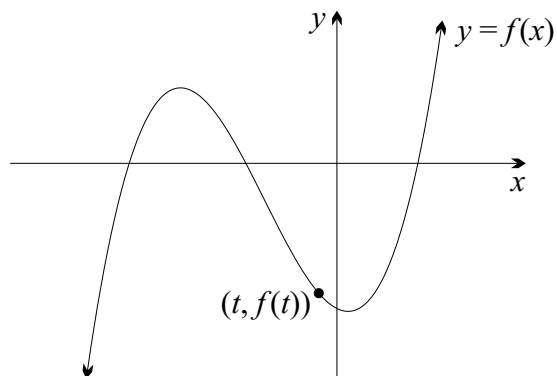
QUESTION THREE

Which of the following is equal to $\frac{1}{\sqrt{5} + 2\sqrt{3}}$?

- (A) $\frac{\sqrt{5} - 2\sqrt{3}}{7}$
- (B) $\frac{2\sqrt{3} + \sqrt{5}}{7}$
- (C) $\frac{2\sqrt{3} - \sqrt{5}}{7}$
- (D) $\frac{\sqrt{5} + 2\sqrt{3}}{-7}$

Exam continues next page ...

QUESTION FOUR



The diagram shows the graph of $y = f(x)$. Which of the following statements is true?

- (A) $f'(t) > 0$ and $f''(t) < 0$
- (B) $f'(t) > 0$ and $f''(t) > 0$
- (C) $f'(t) < 0$ and $f''(t) < 0$
- (D) $f'(t) < 0$ and $f''(t) > 0$

QUESTION FIVE

The acceleration of a particle is given by $\ddot{x} = 4\cos 2t$ where x is the displacement in metres and t is time in seconds. Which of the following is a possible expression for its displacement?

- (A) $-2\sin 2t$
- (B) $2\sin 2t$
- (C) $\cos 2t$
- (D) $-\cos 2t$

QUESTION SIX

Which of the following is the derivative of $y = \frac{e^{7x}}{e^{3x}}$?

- (A) $4e^{4x}$
- (B) e^{4x}
- (C) $\frac{7e^{3x}e^{7x} + 3e^{3x}e^{7x}}{e^{9x}}$
- (D) $\frac{3e^{3x}e^{7x} - 7e^{7x}e^{3x}}{e^{9x}}$

QUESTION SEVEN

A particle moves so that its displacement in metres from the origin at time t seconds is given by $x = 20t - 5t^2$. At what time is it stationary?

- (A) 0 seconds
- (B) 2 seconds
- (C) 4 seconds
- (D) 6 seconds

QUESTION EIGHT

How many terms are in the series $31 + 44 + 57 + \cdots + 226$?

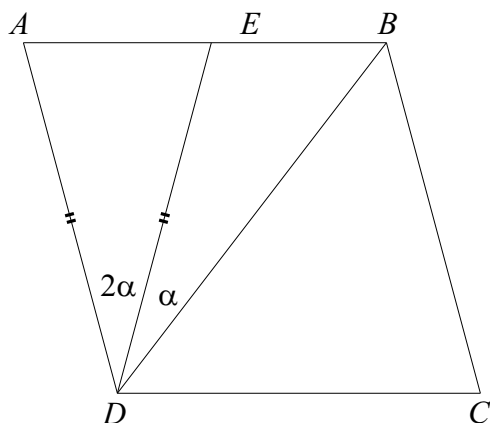
- (A) 4
- (B) 13
- (C) 15
- (D) 16

QUESTION NINE

Given that $\int_0^4 (x + k) dx = 12$ and k is a constant, what is the value of k ?

- (A) 1
- (B) -1
- (C) 0
- (D) 8

QUESTION TEN



The point E lies on the side AB of the rhombus $ABCD$ such that $AD = DE$. The angle ADE is 2α and the angle EDB is α . Find the value of α .

- (A) 45°
- (B) 30°
- (C) 18°
- (D) 15°

————— End of Section I —————

Exam continues overleaf ...

SECTION II - Written Response

Answers for this section should be recorded in the booklets provided.

Show all necessary working.

Start a new booklet for each question.

QUESTION ELEVEN (15 marks) Use a separate writing booklet. **Marks**

(a) Integrate $\frac{3}{x}$ with respect to x .

1

(b) Factorise $3x^2 - 7x + 2$.

2

(c) Solve $\frac{5x - 8}{x} = 1$.

2

(d) Find the equation of the tangent to the curve $y = x^3 + 4$ at the point $(1, 5)$.

2

(e) Differentiate $y = \cos(6x + 5)$.

2

(f) Find the exact value of θ such that $\sin 2\theta = 1$, where $0 \leq \theta \leq \pi$.

2

(g) A sector with radius 5 cm has an arc length of 20 cm. Find the area of the sector.

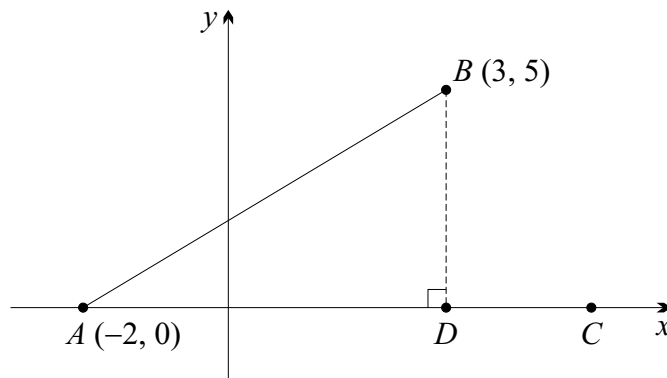
2

(h) Find the limiting sum of the series $\frac{17}{3} + \frac{17}{9} + \frac{17}{27} + \dots$.

2

QUESTION TWELVE (15 marks) Use a separate writing booklet.**Marks**

(a)

**NOT TO SCALE**

The diagram shows the points $A(-2, 0)$, $B(3, 5)$ and the point C which lies on the x -axis. The point D also lies on the x -axis such that BD is perpendicular to AC .

- (i) Show that the gradient of AB is 1.
- (ii) Find the equation of the line AB .
- (iii) What is the size of $\angle BAC$?
- (iv) The length of BC is 13 units. Find the length of DC .
- (v) Calculate the area of $\triangle ABC$.
- (vi) Calculate the size of $\angle ABC$, to the nearest degree.

1**1****1****1****1****2**

- (b) A particle moves on a horizontal line so that its displacement x cm to the right of the origin at time t seconds is given by the function

$$x = \frac{1}{3}t^3 - 6t^2 + 27t - 18.$$

- (i) Find the velocity function.
- (ii) When is the particle stationary?
- (iii) Find the acceleration function.
- (iv) When is the acceleration zero?
- (v) Where is the particle when the acceleration is zero?

1**1****1****1****1**

- (c) A company starts with 60 employees. At the beginning of each subsequent year the number of employees increases by 15%.

- (i) Find a formula for the number of employees at the beginning of the n th year.
- (ii) In which year did the number of employees first exceed 120?

1**2**

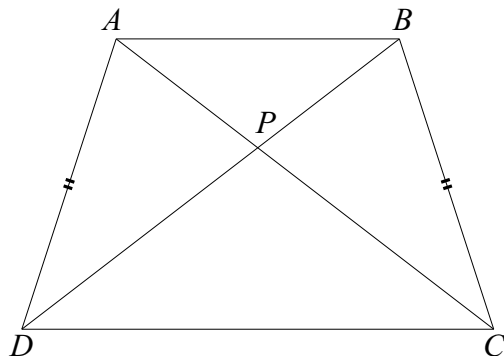
QUESTION THIRTEEN (15 marks) Use a separate writing booklet.

Marks

- (a) (i) Find the values of m for which the equation $mx^2 - 4x + m = 0$ has real roots. 2
- (ii) (α) For what values of m does the equation $mx^2 - 4x + m = 0$ have one root only? 1
- (β) Find this root for each value of m in (α). 1
- (b) The rate of increase in the number of bacteria N in a culture after t hours is proportional to the number present. This can be represented by the differential equation $\frac{dN}{dt} = kN$. Initially there are 1000 bacteria present and two hours later there are 1080.
- (i) Show that $N = 1000e^{kt}$, where k is a constant, is a solution to the differential equation $\frac{dN}{dt} = kN$. 1
- (ii) Find the exact value of k . 1
- (iii) Find the number of bacteria present after a further two hours. 2
- (iv) At what time will the culture have doubled its initial size? 2
- (c) Suppose that $f'(x) = \sin 2x$ and $f(\pi) = 1$.
- (i) Find the function $f(x)$. 3
- (ii) Find the exact value of $f(\frac{\pi}{3})$. 2

QUESTION FOURTEEN (15 marks) Use a separate writing booklet.**Marks**

(a)



The quadrilateral $ABCD$ has diagonals AC and BD which intersect at P .

It is known that $AD = BC$ and $AC = BD$.

Copy the diagram into your answer booklet.

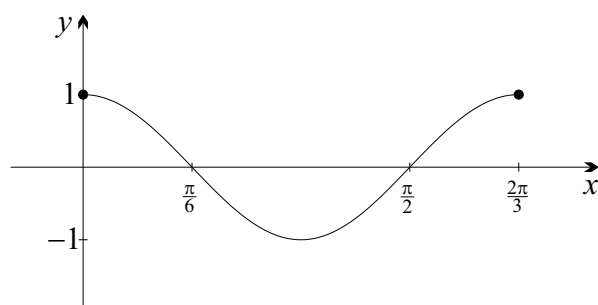
- | | |
|---|----------|
| (i) Prove that the triangles ABC and BAD are congruent. | 3 |
| (ii) Show that triangle ABP is isosceles. | 2 |
| (iii) Hence show that triangle CDP is isosceles. | 2 |
| (iv) Show that AB is parallel to CD . | 3 |
|
(b) | |
| (i) Find the gradient of the tangent to $y = \sin x$ at the origin. | 1 |
| (ii) Draw the graphs of $y = \sin x$, $y = \frac{2}{3}x$ and the tangent in part (i). Draw your three graphs on the same set of axes for $0 \leq x \leq \pi$. | 3 |
| (iii) For what values of m does the equation $\sin x = mx$ have a solution in the domain $0 < x < \pi$? | 1 |

QUESTION FIFTEEN (15 marks) Use a separate writing booklet.

Marks

3

(a)



The diagram above shows the graph of the function $y = \cos 3x$. Find the total area bounded by $y = \cos 3x$ and the x -axis from $x = 0$ to $x = \frac{\pi}{3}$.

(b) If α and β are the roots of the quadratic equation $5x^2 - x - 3 = 0$, find the value of:

(i) $\alpha + \beta$

1

(ii) $\alpha\beta$

1

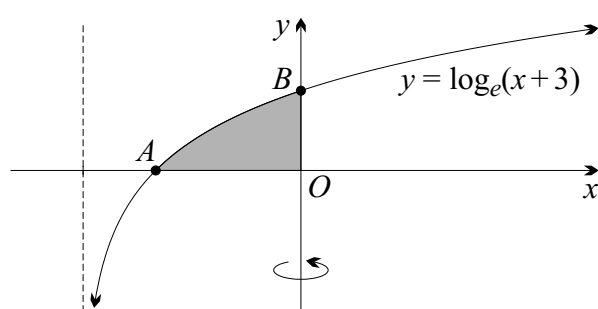
(iii) $\alpha^2 + \beta^2$

1

(iv) $\frac{1}{\alpha} + \frac{1}{\beta}$

1

(c)



The diagram shows the graph of the function $y = \log_e(x + 3)$. The graph crosses the axes at A and B as shown.

(i) Write down the coordinates of B .

1

(ii) Write x as a function of y .

1

(iii) Find the exact value of the volume generated when the shaded region AOB is rotated about the y -axis.

3

QUESTION FIFTEEN (Continued)

(d) Consider the function given by $y = \sin^2 x$.

(i) Copy and complete the following table in your answer booklet.

1

x	0	$\frac{\pi}{4}$	$\frac{\pi}{2}$	$\frac{3\pi}{4}$	π
y					

(ii) Use Simpson's rule with five function values to find an approximation to

2

$$\int_0^{\pi} \sin^2 x \, dx.$$

The Exam continues on the next page

Exam continues overleaf ...

QUESTION SIXTEEN (15 marks) Use a separate writing booklet.**Marks**

- (a) A company borrows \$800 000 to update its car fleet. The interest rate is 12% p.a. compounded monthly. It pays off the loan by 24 equal monthly instalments. The first instalment is paid one month after the loan is taken out.

Let A_n be the amount owing after n instalments are paid. Let M be the amount of each instalment.

- (i) Show that the amount owing after two months is $A_2 = 816\,080 - M(2.01)$. 2

- (ii) Show that $M = \frac{8000 \times 1.01^{24}}{1.01^{24} - 1}$. 2

- (iii) Hence calculate M to the nearest dollar. 1

- (iv) After paying ten instalments, the company decides to increase its repayments to \$60 000 each month. Find the total number of months it takes the company to pay off its debt. 3

- (b) A van is to travel 1000 kilometres at a constant speed of v km/h.

When travelling at v km/h, the van uses fuel at a rate of $(6 + \frac{v^2}{50})$ litres per hour.

The truck company pays \$1.50 per litre for fuel and pays each of the two drivers \$30 per hour while the van is travelling.

- (i) Let the total cost of fuel and the drivers' wages for the trip be C dollars. Show that 3

$$C = \frac{69\,000}{v} + 30v.$$

- (ii) The van must take no longer than 12 hours to complete the trip, and speed limits require that $v \leq 110$. 4

At what speed v should the van travel to minimise the cost C ?

_____ End of Section II _____

END OF EXAMINATION

B L A N K P A G E

The following list of standard integrals may be used:

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE : $\ln x = \log_e x, \quad x > 0$



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- Record your multiple choice answers by filling in the circle corresponding to your choice for each question.
- Fill in the circle completely.
- Each question has only one correct answer.

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CANDIDATE NUMBER

Question One

A ☐ B ☐ C ☐ D ☐

Question Two

A ☐ B ☐ C ☐ D ☐

Question Three

A ☐ B ☐ C ☐ D ☐

Question Four

A ☐ B ☐ C ☐ D ☐

Question Five

A ☐ B ☐ C ☐ D ☐

Question Six

A ☐ B ☐ C ☐ D ☐

Question Seven

A ☐ B ☐ C ☐ D ☐

Question Eight

A ☐ B ☐ C ☐ D ☐

Question Nine

A ☐ B ☐ C ☐ D ☐

Question Ten

A ☐ B ☐ C ☐ D ☐



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CANDIDATE NUMBER:

Question One

A ☐ B ☒ C ☐ D ☐

Question Two

A ☐ B ☐ C ☒ D ☐

Question Three

A ☐ B ☐ C ☒ D ☐

Question Four

A ☐ B ☐ C ☐ D ☒

Question Five

A ☐ B ☐ C ☐ D ☒

Question Six

A ☒ B ☐ C ☐ D ☐

Question Seven

A ☐ B ☒ C ☐ D ☐

Question Eight

A ☐ B ☐ C ☐ D ☒

Question Nine

A ☒ B ☐ C ☐ D ☐

Question Ten

A ☐ B ☐ C ☒ D ☐

Solutions

20 Trial 2014

1. -2 B

2. 5.30 C

3. $(\sqrt{5} + 2\sqrt{3})(\sqrt{5} - 2\sqrt{3}) = 5 - 12 = -7$
 $\frac{\sqrt{5} - 2\sqrt{3}}{-7}$ or $\frac{2\sqrt{3} - \sqrt{5}}{7}$

C

4. D D

5. D

6. $y = e^{7x} \cdot e^{-3x}$
 $= e^{4x}$
 $y' = 4e^{4x}$

A

7. $x = 20t - 5t^2$
 $\dot{x} = 20 - 10t = 0$
 $t = 2$

B

8. AP $a = 31$, $d = 13$ $T_n = 226$

$$T_n = a + (n-1)d$$

$$226 = 31 + (n-1)13$$

$$= 31 + 13n - 13$$

$$226 = 18 + 13n$$

$$13n = 208$$

$$n = 16$$

D

9. $\int_0^4 (x+k) dx = 12.$

$$\left[\frac{x^2}{2} + kx \right]_0^4 = 12$$

$$\frac{16}{2} + 4k = 12$$

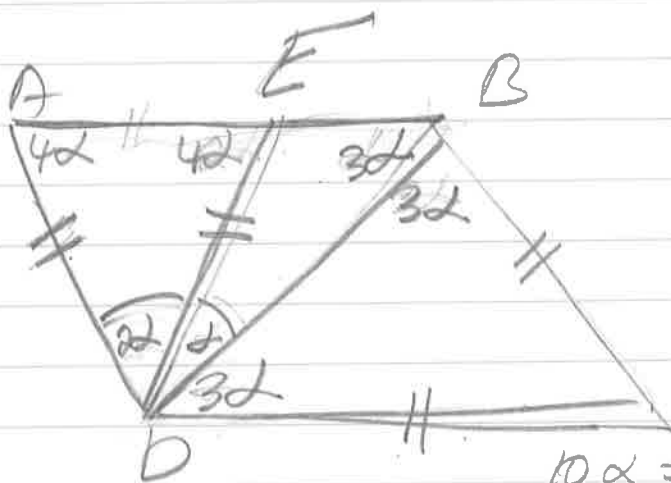
$$4k = 12 - 8$$

$$= 4$$

$$k = 1$$

A

10.



$$10x = 180^\circ$$

$$x = 18^\circ$$

C

11.

$$a) \int \frac{3}{x} dx = 3 \log_e x + C \quad \checkmark \quad (C \text{ not required})$$

$$b) 3x^2 - 2x + 2 = (3x - 1)(x - 2) \quad \checkmark \checkmark$$

$$c) \begin{aligned} 5x - 8 &= 2x & \checkmark \\ 4x &= 8 \\ x &= 2 & \checkmark \end{aligned}$$

$$d) \begin{aligned} y &= x^3 + 4 \\ \frac{dy}{dx} &= 3x^2 & \checkmark \end{aligned}$$

$$x=1, \quad m=3 \quad \checkmark$$

$$y - y_1 = m(x - x_1)$$

$$y - 5 = 3(x - 1) \quad \checkmark$$

$$y - 5 = 3x - 3$$

$$y = 3x + 2$$

$$e) \begin{aligned} y &= \cos(6x + 5) \\ \frac{dy}{dx} &= -6 \sin(6x + 5) \end{aligned} \quad \checkmark \checkmark$$

$$f) \begin{aligned} \sin 2\theta &= 1 \\ 2\theta &= \frac{\pi}{2} & \checkmark \\ \theta &= \frac{\pi}{4} & \checkmark \end{aligned}$$

$$g) \begin{aligned} \ell &= r\theta \\ 20 &= 5\theta & \checkmark \end{aligned}$$

$$\begin{aligned} A &= \frac{1}{2} r^2 \theta, \quad (r\theta = 20, \quad r = 5) \\ &= \frac{1}{2} \times 5 \times 20 \\ &= 50 \text{ cm}^2 \end{aligned} \quad \checkmark$$

(ignore units)

$$h) \quad \frac{12}{3} + \frac{12}{9} + \frac{12}{27} + \dots$$

$$GP, \quad a = \frac{12}{3}, \quad r = \frac{1}{3} \quad \checkmark$$

$$S_{\infty} = \frac{a}{1-r}$$

$$= \frac{\frac{12}{3}}{1-\frac{1}{3}}$$

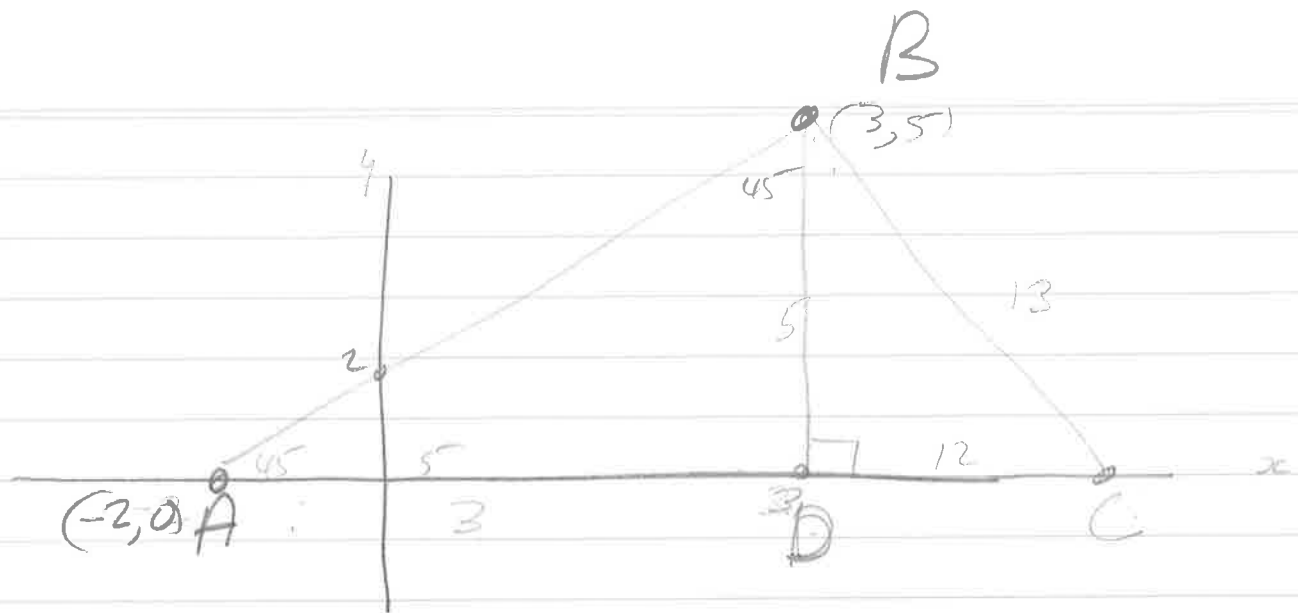
$$= \frac{\frac{12}{3}}{\frac{2}{3}}$$

$$= \frac{12}{3} \times \frac{3}{2}$$

$$= \frac{12}{2} \quad \checkmark$$

12.

a)



i)
$$m = \frac{y_1 - y_2}{x_1 - x_2}$$

$$= \frac{5 - 0}{3 - (-2)}$$

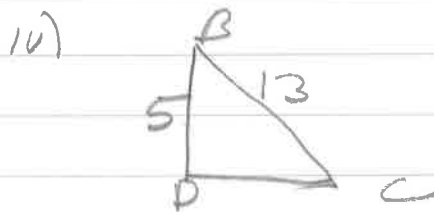
$$= 1$$

ii)
$$y - y_1 = m(x - x_1)$$

$$y - 0 = 1(x + 2)$$

$$y = x + 2$$

iii) $\angle BAC = 45^\circ$



$$DC^2 = 13^2 - 5^2$$

$$DC = 12$$

iv) $|AC| = 2 + 3 + 12 = 17$

area
$$= \frac{1}{2} \times 17 \times 5$$

$$= 42.5$$

b)

vi) $\sin \angle DBC = \frac{12}{13}$

$\angle DBC = 67^\circ 23' \quad \checkmark$

$\angle ABD = 45^\circ$

so $\angle ABC = 45^\circ + 67^\circ 23'$
 $= 112^\circ 23'$
 $\approx 112^\circ \quad \checkmark$

OR use the sine rule.

b) $x = \frac{1}{3}t^3 - 6t^2 + 27t - 18$

(i) $\dot{x} = t^2 - 12t + 27 \quad \checkmark$

(ii) $t^2 - 12t + 27 = 0$

$(t - 3)(t - 9) = 0$

$t = 3 \text{ and } 9 \text{ sec} \quad \checkmark$

(iii) $\ddot{x} = 2t - 12$

(iv) $t = 6 \text{ seconds}$

(v) $x = \frac{1}{3} \times 6^3 - 6 \times 6^2 + 27 \times 6 - 18$

$= 0 \text{ cm.}$

it is at the origin $\checkmark$

c) (i) $M_1 = 60(1.15)$

$M_n = 60(1.15)^{n-1} \quad \checkmark$

(ii) $60(1.15)^{n-1} = 120$

$(1.15)^{n-1} = 2 \quad \checkmark$

$$60$$

$$60(1.15)$$

$$60(1.15)^2$$

$$60(1.15)^3$$

$$60(1.15)^4$$

$$60(1.15)^5$$

120.68.

$$(n-1) \log 1.15 = \log 2$$

$$n \log 1.15 - \log 1.15 = \log 2$$

$$n \log 1.15 = \log 2 + \log 1.15$$

$$n = \frac{\log 2 + \log 1.15}{\log 1.15}$$

$$= 5.959 \dots$$

(OR)

Guess/Check

During first year	60
" 2nd yr	$60(1.15)$
3rd yr	$60(1.15)^2$
4th yr	$60(1.15)^3$
5th yr	$60(1.15)^4 = 104.94$
6th yr	$60(1.15)^5 = 120.68$



The number of employees exceeds 120 in the 6th year.

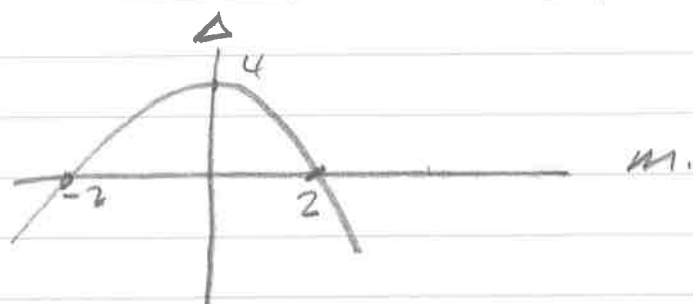


Q13.

a) (i) $mx^2 - 4x + m = 0$
for real roots $\Delta \geq 0$.

$$\Delta = 16 - 4m^2 \geq 0 \quad \checkmark$$

$$4(4 - m^2) \geq 0.$$



$$\Delta \geq 0 \text{ for } -2 \leq m \leq 2. \quad \checkmark$$

(ii) (a) $mx^2 - 4x + m = 0$.
for one root, $\Delta = 0$.

also include
sols in m &
solve eqn.

$$16 - 4m^2 = 0$$

$$m = 2 \text{ or } -2. \quad \checkmark$$

(b) $x = \frac{-b}{2a}$
 $= \frac{4}{2m}$

$$m = 2, \quad x = \frac{4}{4} = 1$$

$$m = -2, \quad x = \frac{4}{-4} = -1$$

} $\checkmark$ need both.

$$b) \quad \frac{dN}{dt} = kN$$

$$t=0, N_0 = 1000$$

$$t=2, N_2 = 1080.$$

$$(i) \quad \left. \begin{aligned} N &= 1000 e^{kt} \\ \frac{dN}{dt} &= k \cdot 1000 e^{kt} \\ &= kN \end{aligned} \right\} \checkmark$$

$$(ii) \quad N = 1000 e^{kt}$$

$$t=2, \quad 1080 = 1000 e^{2k}$$

$$e^{2k} = 1.08$$

$$\text{or } e^{2k} = \frac{27}{25}$$

$$\checkmark \quad 2k = \ln 1.08$$

$$2k = \frac{27}{25}$$

$$k = \frac{1}{2} \ln 1.08$$

$$k = \frac{1}{2} \ln \frac{27}{25}$$

$$(iii) \quad t=4, \quad N = 1000 e^{2 \ln 1.08} \quad \checkmark$$

$$= 1000 \times 1.08^2$$

$$= 1166.4$$

$$\approx 1166 \quad \checkmark$$

$$iv) \quad \text{find } t \text{ when } N = 2000$$

$$2000 = 1000 e^{kt}$$

$$2 = e^{kt}$$

$$kt = \ln 2 \quad \checkmark$$

$$t = \ln 2 \div \frac{1}{2} \ln 1.08$$

≈ 18 hours after initial time. ✓

(c) $f'(x) = \sin 2x$, $f(\pi) = 1$.

(i) $f(x) = \int \sin 2x \, dx$

$$= -\frac{1}{2} \cos 2x + C \quad \checkmark$$

$x = \pi$, $1 = -\frac{1}{2} \cos 2\pi + C$

$$1 = -\frac{1}{2} + C \quad \text{so} \quad C = \frac{3}{2} \quad \checkmark$$

$$f(x) = -\frac{1}{2} \cos 2x + \frac{3}{2} \quad \checkmark$$

(ii) $f\left(\frac{\pi}{3}\right) = -\frac{1}{2} \cos \frac{2\pi}{3} + \frac{3}{2}$

$$= -\frac{1}{2} \times \left(-\frac{1}{2}\right) + \frac{3}{2} \quad \checkmark$$

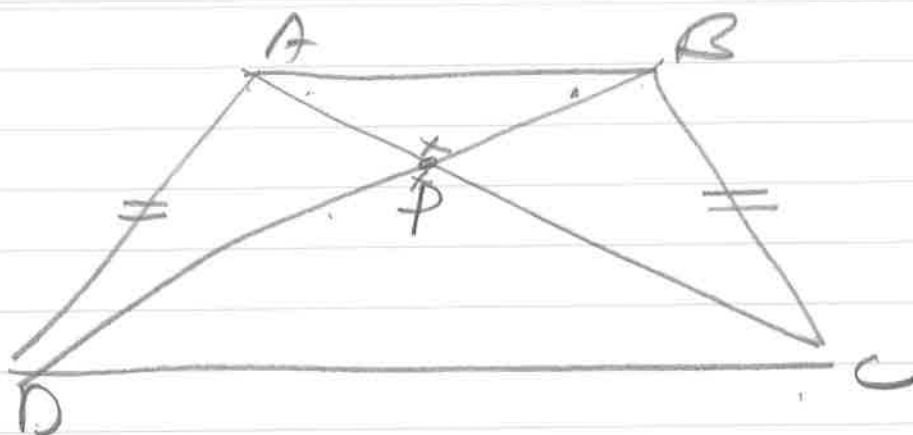
$$= \frac{1}{4} + \frac{3}{2}$$

$$= \frac{7}{4} \quad \checkmark$$



Q 14.

a)



(i) In $\triangle ABC$, $\triangle BAD$.

AB is common

$BC = AD$, given

$AC = BD$ given

$\therefore \triangle ABC \equiv \triangle BAD$, SSS

✓ ✓

✓ need reason

(iii) Matching angles in congruent triangles are equal ✓

$\therefore \angle ABD = \angle DAB$

So $\triangle APB$ is isosceles since it has two equal angles. ✓

iii) Now $AC = DB$ given

$AP = BP$, sides opposite equal angles in $\triangle APB$. ✓

So $AC - AP = BD - BP$.

i.e. $PD = PC$ and $\triangle DPC$ is isosceles. ✓

(can also show $\angle D = \angle C$).

iv) $\angle APB = \angle DPC$, vertically opposite

So $\angle PAB + \angle ABP = \angle PDC + \angle PCD$, angle

sum of $\triangle APB$, $\triangle DPC$ are 180° ✓

But both are isosceles.

So $\angle ABP = \angle PDC$ ✓

But these are alternate } ✓

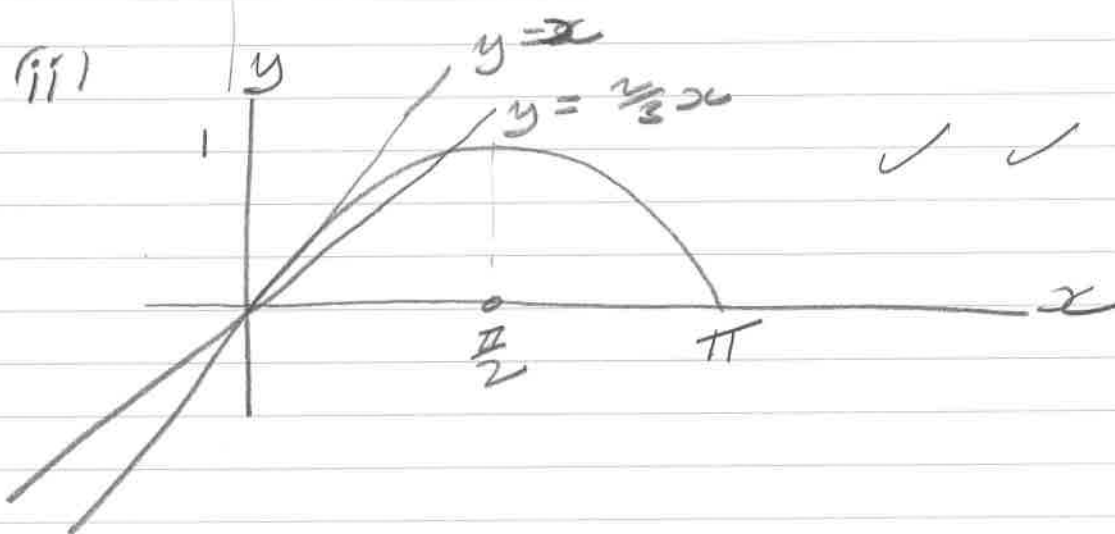
So $AB \parallel DC$

(b) (i) $y = \sin x$
 $\frac{dy}{dx} = \cos x$

$x=0$, $m = \cos 0$
 so $m = 1$



(can just state $m=1$).



(iii) $0 < m < 1$.

Q15

$$a) \quad A = 2 \int_0^{\frac{\pi}{6}} \cos 3x dx \quad \checkmark$$

$$= \frac{2}{3} \left[\sin 3x \right]_0^{\frac{\pi}{6}} \quad \checkmark$$

$$= \frac{2}{3} (\sin \frac{\pi}{2} - \sin 0)$$

$$= \frac{2}{3} \times 1 \quad \checkmark$$

$$b) \quad 5x^2 - x + 3 = 0$$

$$(i) \quad \alpha + \beta = \frac{1}{5} \quad \checkmark$$

$$(ii) \quad \alpha\beta = -\frac{3}{5} \quad \checkmark$$

$$(iii) \quad \alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta$$

$$= \frac{1}{25} + \frac{6}{5}$$

$$= \frac{31}{25} \quad \checkmark$$

$$\begin{aligned}
 (v) \quad \frac{1}{\alpha} + \frac{1}{\beta} &= \frac{\alpha + \beta}{\alpha\beta} \\
 &= \frac{\frac{1}{5}}{-\frac{3}{5}} \\
 &= \frac{1}{5} \times -\frac{5}{3} \\
 &= -\frac{1}{3}.
 \end{aligned}$$



(c) (i) B is $(0, \ln 3)$.



(ii) $y = \ln(x+3)$
 $e^y = x+3$
 $x = e^y - 3$



(iii) $V = \pi \int_0^{\ln 3} x^2 dy$

$$= \pi \int_0^{\ln 3} (e^y - 3)^2 dy$$



$$= \pi \int_0^{\ln 3} e^{2y} - 6e^y + 9 dy$$

$$= \pi \left[\frac{1}{2} e^{2y} - 6e^y + 9y \right]_0^{\ln 3}$$



$$= \pi \left[\left(\frac{1}{2} \times 9 - 6 \times 3 + 9 \ln 3 \right) - \left(\frac{1}{2} - 6 + 0 \right) \right]$$

$$= \pi \left(4\frac{1}{2} - 18 + 9 \ln 3 + 5\frac{1}{2} \right)$$

$$= \pi (-8 + 9 \ln 3)$$



(d)

$$y = \sin^2 x.$$

$$\sin \frac{3\pi}{4}$$

(ii)

x	0	$\frac{\pi}{4}$	$\frac{\pi}{2}$	$\frac{3\pi}{4}$	π
y	0	$\frac{1}{2}$	1	$\frac{1}{2}$	0

✓

(iii)

$$\int_0^{\pi} \sin^2 x dx = \frac{\pi}{2} \left(0 + \frac{1}{2} + 1 + 1 + \frac{1}{2} + 0 \right) \checkmark$$

$$= \frac{\pi}{2} \times 6$$

$$= \frac{\pi}{2} \quad \checkmark$$

Q16

(a)

$$i) A_1 = 800\,000(1.01) - M$$

$$A_2 = 800\,000(1.01)^2 - M(1.01) - M$$

$$= 800\,000(1.01)^2 - M(1.01 + 1)$$

$$= 800\,000(1.01)^2 - M(2.01)$$

(i)

$$A_{24} = 800\,000(1.01)^{24} - M(1.01^{23} + 1.01^{22} + \dots + 1)$$

when loan is paid off $M=0$

$$\text{so } 800\,000(1.01)^{24} - M(1.01^{23} + 1.01^{22} + \dots + 1) = 0$$

• need to show GP for this m.k.

$$800\,000(1.01)^{24} = \frac{M(1.01^{24} - 1)}{0.01}$$

✓ need to see
if $A_{24}=0$ or some
variation of it.

$$M = \frac{800\,000(1.01)^{24} \times 0.01}{(1.01)^{24} - 1}$$

$$= \frac{8000(1.01)^{24}}{(1.01)^{24} - 1}$$

iii)

$$M \approx \$37\,659$$

$$iv). A_{10} = 800\,000(1.01)^{10} - 37659 \left(\frac{1.01^{10} - 1}{0.01} \right) \\ = \$489701 \quad \checkmark$$

So we need to pay this off with
 $M = \$60\,000$

$$0 = 489701(1.01)^n - 60\,000 \left(\frac{1.01^n - 1}{0.01} \right) \quad \checkmark$$

$$489701(1.01)^n = 6000000(1.01^n) - 6000000$$

$$(1.01)^n (6000000 - 489701) = 6000000$$

$$(1.01)^n = \frac{6000000}{5510299}$$

$$= 1.08887$$

$$n = \frac{\ln(1.08887)}{\ln 1.01}$$

$$= 8.5$$

$$\approx 9 \text{ months} \quad \checkmark$$

It takes 19 months to pay off
the debt.

16

(b) (i) Time for trip = $\frac{1000}{v}$ hrs ✓

Drivers' wages = $\frac{30 \times 1000}{v} \times 2$ ✓

No of litres of fuel = $\$1.50 \times \frac{1000}{v} \times \left(6 \times \frac{v^2}{50}\right)$
 $= \frac{1500}{v} \times \left(6 \times \frac{v^2}{50}\right)$ ✓

$= \frac{9000}{v} + 30v$

So Total = $\frac{60000}{v} + \frac{9000}{v} + 30v$

$C = \frac{69000}{v} + 30v$ dollars.

(ii) $C = 69000v^{-1} + 30v$

$\frac{dC}{dv} = -69000v^{-2} + 30$

$= 0$ at stationary point

$\frac{69000}{v^2} = 30$

$v^2 = 2300$

$v = \sqrt{2300}$
 ≈ 48 km/hr. ✓

Now if $v = \sqrt{2300}$, then $t = \frac{1000}{\sqrt{2300}}$ hrs
 ≈ 20.8 hours.

This is too long, we must take no longer than 12 hours. ✓

We want $\frac{1000}{v} \leq 12$

$$v \geq \frac{1000}{12}$$

$$v \geq 83\frac{1}{3} \text{ km/hr} \quad \checkmark$$

$$\text{if } v = 83\frac{1}{2}, \quad C = \frac{69000}{83\frac{1}{2}} + 30 \times 83\frac{1}{2} \\ = \$3328$$

$$v = 110, \quad C = \frac{69000}{110} + 30 \times 110 \\ = \$3427$$

check concavity: $\frac{d^2C}{dv^2} = 2 \times 69000 v^{-3} > 0$
for all $v > 0$

So $v = 48$ km/hr gives a minimum turning point, and C is increasing after $v = 48$. As v increases, C increases, so, the minimum C that satisfies the conditions is $v = 83\frac{1}{3}$ km/hr.